

Project Handoff: Primitive Cognitive Patterns of Mathematics (PCPM)

Overview

This project aims to investigate whether the Grade 10 mathematics curriculum can be reduced to a significantly smaller set of **primitive cognitive patterns**.

Rather than organizing mathematics by traditional topics (e.g., algebra, geometry, exponents, systems of equations), this project proposes organizing it by the underlying reasoning required to solve problems.

The hypothesis is that hundreds of individual curriculum skills may be expressions of a relatively small number of reusable cognitive patterns.

Research Goal

Primary Objective

Discover the smallest set of primitive cognitive patterns capable of classifying and supporting the solution of every Grade 10 mathematics skill.

The emphasis is on the **reasoning process**, not on replacing mathematical content.

Research Hypothesis

Current mathematics curricula are organized around subject domains.

Examples include:

- Algebra
- Geometry
- Functions
- Exponents
- Radicals
- Trigonometry
- Probability

This project proposes an alternative hypothesis:

These domains may be compositions of a much smaller set of primitive reasoning operations.

If true, students could first learn these primitive patterns and then learn to recognize which pattern (or combination of patterns) a problem requires.

Scientific Method

The project will follow a bottom-up approach.

We will **not** invent primitives first.

Instead, we will derive them from the curriculum.

Step 1

Collect every Grade 10 learning objective or skill.

The IXL Grade 10 Mathematics curriculum provides a fixed and comprehensive list of skills and serves as the initial dataset.

Each skill should be treated as an independent observation.

Step 2

Ignore the curriculum category.

Do not classify skills by chapter.

Instead ask:

What reasoning operation is the student performing?

Examples:

- Simplify radicals
- Solve systems of equations
- Convert scientific notation
- Factor trinomials
- Find slope
- Determine congruence

Each should initially receive a plain-language description of the reasoning involved.

Step 3

Cluster similar reasoning.

Merge skills that require the same cognitive operation.

Example:

- Scientific notation
- Graphs
- Coordinate representation
- Word problems

may all involve:

Represent

Another cluster may include:

- Factoring
- Polynomial decomposition
- Geometric decomposition

which may become:

Decompose

Step 4

Continue merging until no additional meaningful reductions are possible.

The remaining set becomes the candidate primitive pattern language.

Candidate Primitive Pattern Requirements

Every primitive should satisfy the following criteria.

1. General

A primitive must apply to many different skills.

Example:

✗ Solve Quadratic Equations

✓ Transform Representation

2. Independent

A primitive should not substantially overlap another primitive.

If two primitives always occur together, they should probably be merged.

3. Complete

Every Grade 10 skill should map to one or, where necessary, two primitives.

If a skill cannot be classified, either:

- a primitive is missing, or
 - the current primitives require refinement.
-

Candidate Primitive Set (Working Draft)

The following list is an initial hypothesis only.

It will evolve during classification.

- Identify
- Represent
- Transform
- Compare
- Compose
- Decompose
- Constrain
- Measure
- Generalize
- Verify

These should be considered placeholders until the classification process is complete.

Classification Process

Every curriculum skill should be transformed into a record.

Example:

Skill:
Simplify Radical Expressions

Traditional Category:
Algebra

Reasoning:
Rewrite an equivalent representation.

Candidate Primitive:
Transform

Another example:

Skill:
Solve Systems of Equations

Traditional Category:
Algebra

Reasoning:
Find values satisfying multiple simultaneous constraints.

Candidate Primitive:
Constrain

Another:

Skill:
Factor Trinomials

Traditional Category:
Algebra

Reasoning:
Break a structure into simpler components.

Candidate Primitive:
Decompose

Validation

The hypothesis is only considered successful if:

- Every Grade 10 skill can be classified.
 - The primitive vocabulary remains relatively small.
 - Different reviewers classify the same skills similarly.
 - The primitive set provides explanatory value beyond the traditional curriculum structure.
-

Deliverables

The project should produce:

1. Primitive Pattern Vocabulary

A concise reference defining each primitive.

2. Classification Matrix

A table mapping every Grade 10 skill to its corresponding primitive(s).

Curriculum Skill	Primitive(s)	Notes
------------------	--------------	-------

3. Pattern Definitions

Each primitive should include:

- Definition
 - Purpose
 - Recognition cues
 - Common transformations
 - Example problems
 - Related primitives
-

4. Recognition Guide

A practical guide that helps students identify which primitive applies before attempting a solution.

Example:

What is the problem asking me to do?

- Represent?
- Transform?
- Compare?
- Decompose?
- Measure?
- Constrain?

This becomes the student's first decision rather than selecting a memorized procedure.

Long-Term Vision

If the hypothesis is supported, the same methodology can be extended beyond Grade 10.

Potential future applications include:

- Grade 11 Mathematics
- Grade 12 Mathematics
- University Mathematics
- Physics
- Computer Science
- Logic
- Engineering

Ultimately, the goal is to investigate whether mathematical education can be viewed as learning a compact set of reusable reasoning primitives instead of hundreds of isolated procedures.

Guiding Principle

The purpose of this research is **not** to simplify mathematics by removing rigor.

The purpose is to determine whether the reasoning underlying mathematics can be expressed through a smaller, reusable vocabulary of cognitive patterns.

If successful, students would learn to recognize *how to think about a problem* before learning *which procedure to apply*.